

Question

When children reason counterfactually, do they prefer the “closest possibilities” for both forward and backward counterfactuals?

Background

Counterfactuals (CFs)

If the bread had burned, the smoke detector *would have* sounded.

Backward CFs

Reasoning backwards, from effect to cause:

- Effect:** If the smoke detector had sounded...
- Cause:** my bread would have burned!



Closest Possibilities

Adults generally tend to reason over counterfactual possibilities which are least different from what had happened, from *the Actual World*. [1]

E.g., “If the bread had burned, the smoke detector would have exploded!” ... Unlikely.

So Far...

Children:

- Given a clear physical causal system (scenario BOTH), 4-year-olds prefer the *closest possible world* for forward CFs [2].
- No prior work on backward CFs...

Adults:

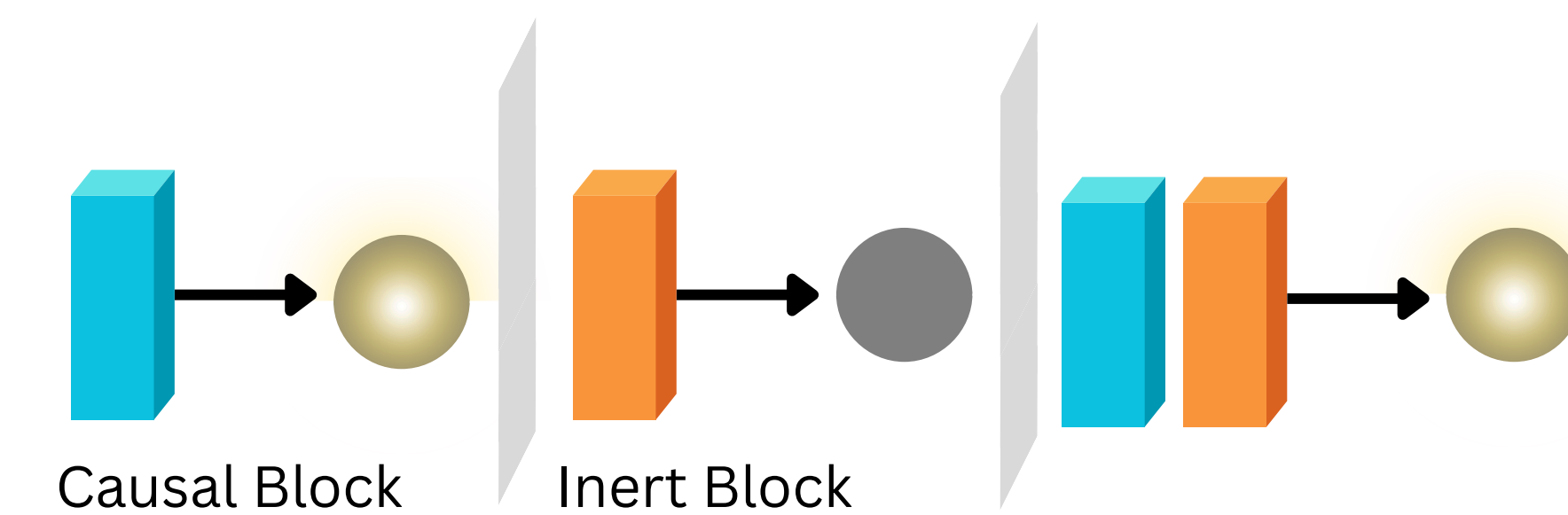
- Using the same method, adults prefer the closest possibilities when CF reasoning backwards [3].
- Other studies have found inconsistent patterning [4].

Method

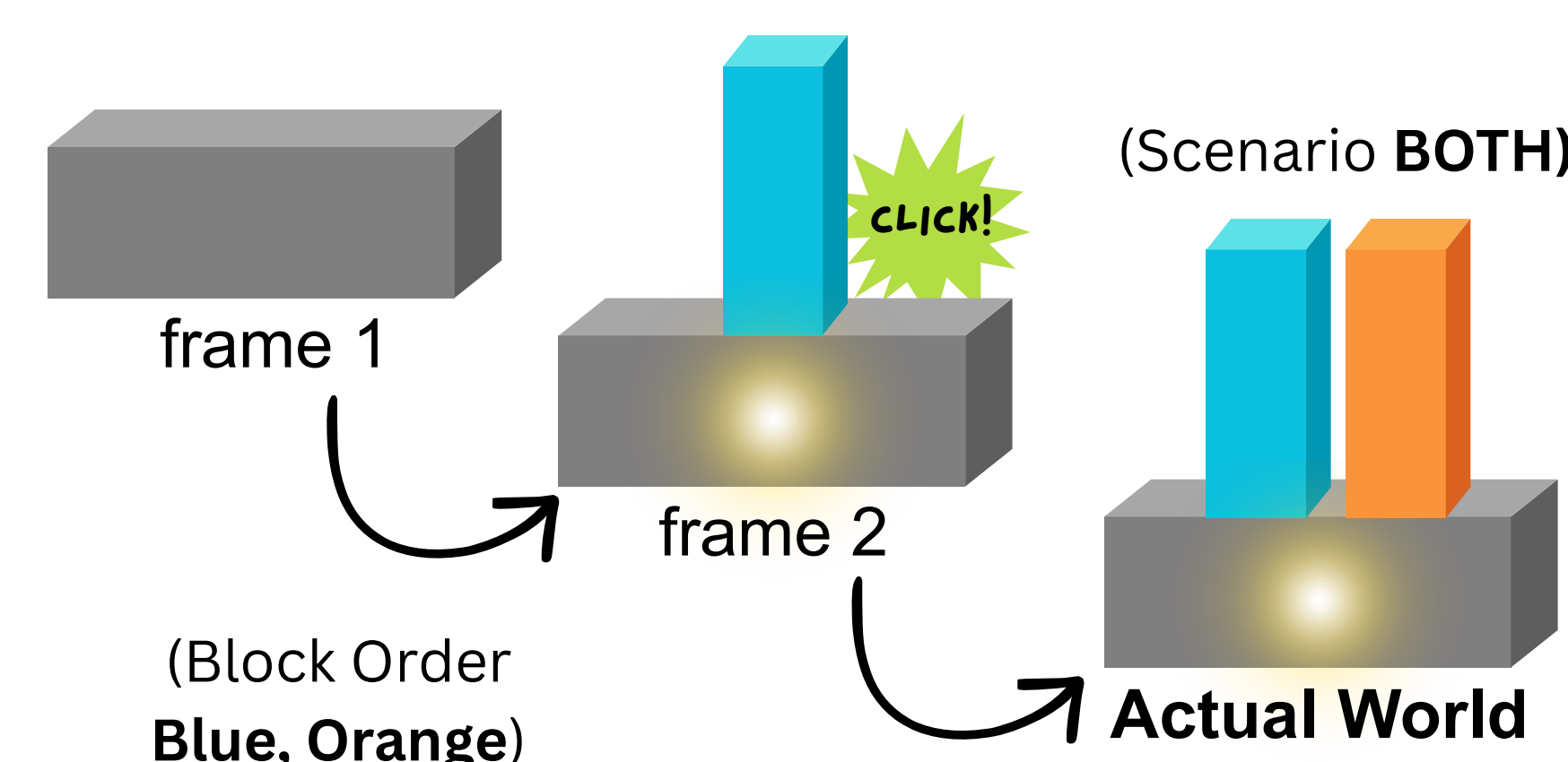
Participants: 31 6-year-olds, 30 7-year-olds; 19 adults.

Procedure:

1. Training Demonstration



3. Test Demonstration



2. Causal Understanding Questions

E.g., “Did the blue block alone make the light turn on?” (Yes)

4. Counterfactual Questions*

FORWARD: “If the orange one had not been on the box, would the light have turned on?” (Yes)

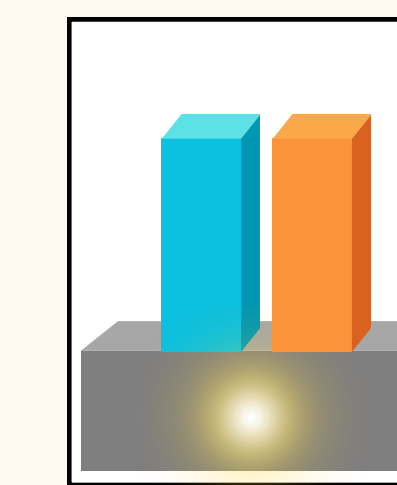
BACKWARD: “If the light had not turned on, would the orange one have been on the box?” (Yes)

*counterbalance between “on” and “off” questioning, where the expected response flips.

Scenarios

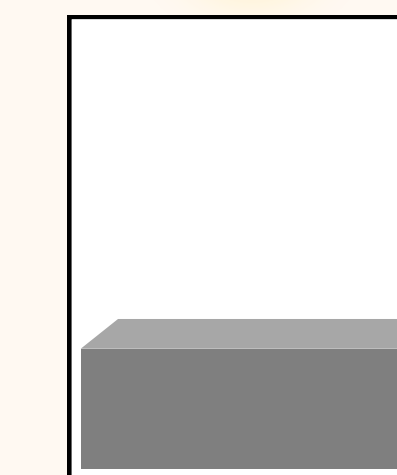
BOTH

- Causal on
- Inert on
- Light on



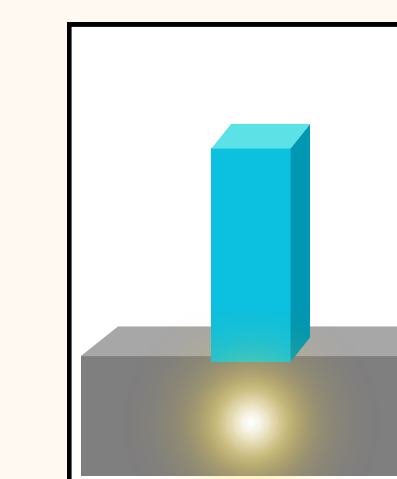
NEITHER

- Causal off
- Inert off
- Light off



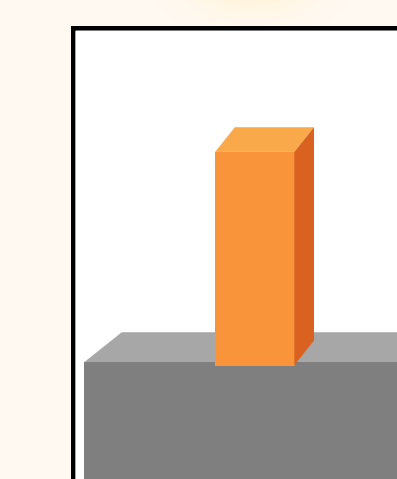
CAUSAL ONLY

- Causal on
- Inert off
- Light on



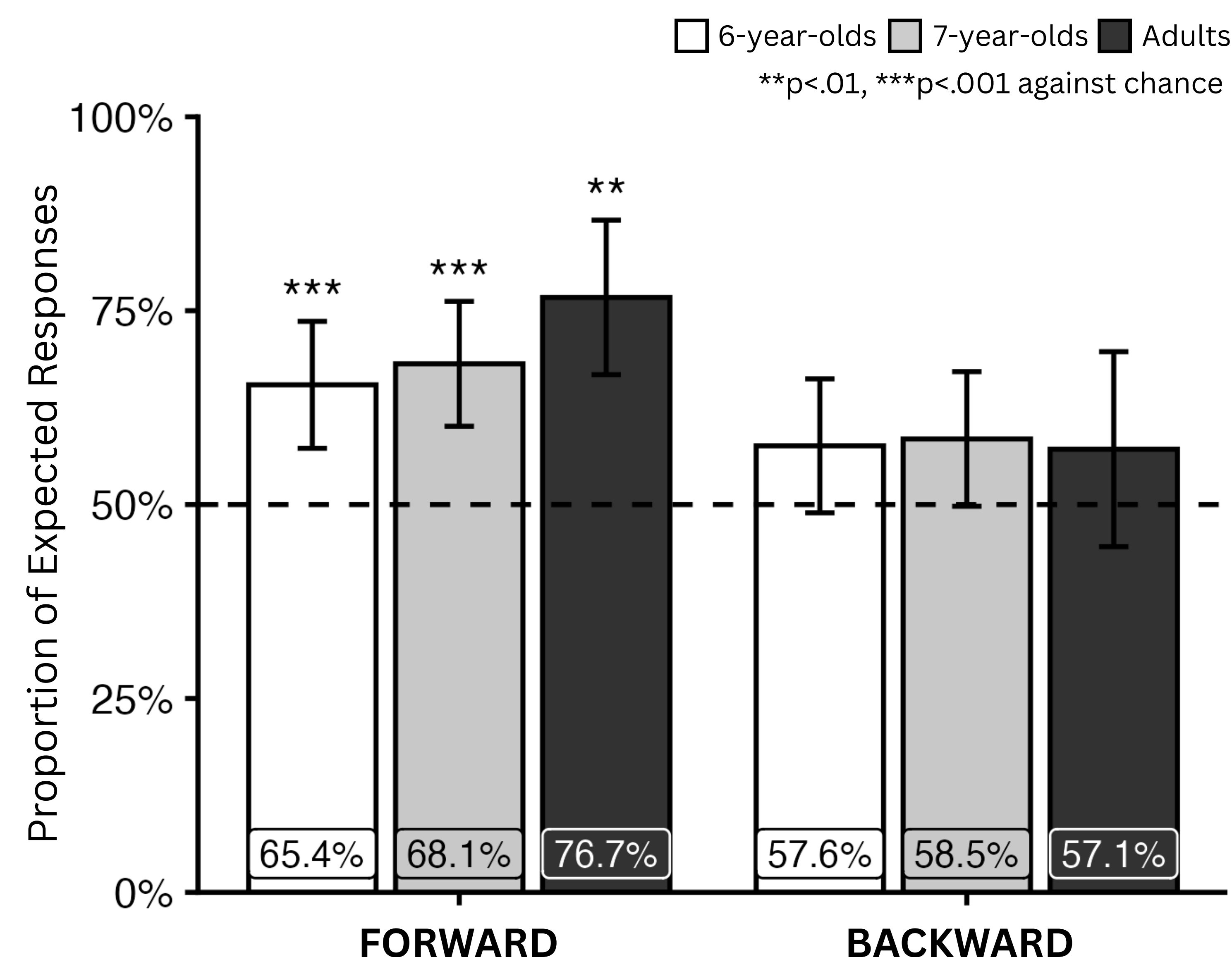
INERT ONLY

- Causal off
- Inert on
- Light off



Results

Responses by Age and Counterfactual Condition



Proportion of Expected Responses across Scenarios

Scenario	Children	Adults
BOTH	73.9%	75.0%
NEITHER	78.8%	78.8%
CAUSAL ONLY	40.7%	71.4%
INERT ONLY	56.8%	43.8%

Note: We expected participants to not change the orange block from the Actual World.

Results

Overall: There was no significant effect of Direction, Age, or Block Order (all ps >.05, GLMM in R).

Chance Comparison:

- FORWARD:** Both 6-and 7-yos responses were above chance.
- BACKWARD:** Both 6-and 7-yos responses were at chance.

Adults:

- Adults were above chance for FORWARD but at chance for BACKWARD.

Scenario:

- CAUSAL ONLY was significantly more difficult than BOTH (p = .015).
- Differences between INERT ONLY and BOTH were greater in BACKWARD than FORWARD (p = .011).

Takeaway

6- and 7-year-olds and adults consistently choose the closest possible world when reasoning counterfactually, only for forwards counterfactuals.

References

[1] Lewis, D. (1973). Counterfactuals and comparative possibility. *Ifs: Conditionals, belief, decision, chance and time*. [2] Nyhout, A. & Ganea, P.A. (2019). Mature counterfactual reasoning in 4- and 5-year-olds. *Cognition*. [3] Grosu, I., Le, D., & Ganea, P. (2025). Backwards counterfactuals and the closest possible world. *Proceedings of the Annual Meeting of the Cognitive Science Society*. [4] Rips, L. (2010). Two Causal Theories of Counterfactual Conditionals. *Cognitive science*. 34. 175-221. 10.1111/j.1551-6709.2009.01080.x.

Acknowledgements

Research supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) allocated to Patricia A. Ganea.